The Geometric Origin of Quantum Mechanics: The Reduced Planck Constant = Action per Radian

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Abstract

Within the framework of the Unified Field Theory, this paper investigates the geometric origin and logical necessity of quantum mechanics. The Unified Field Theory holds that space is not a static background, but flows in a right-handed helical pattern at the composite speed of light c. Under this geometric configuration, a rotational scaling length ℓ_Ω and a helical divergence strength Q are introduced to establish the dynamic scale of space.

Furthermore, through the space–mass coupling relations

$$M = \mu_g \ell_\Omega,$$

$$G = \frac{4\pi c^2}{\mu_g},$$

the geometric origin of mass and gravity is revealed. Mass is not an intrinsic property, but rather a manifestation of the coupling strength between space and an object; likewise, the universal gravitational constant G is no longer an empirical parameter, but naturally derived from the speed of light c and the space–mass coupling constant μ_g.

On this basis, the reduced Planck constant is defined as

$$\hbar = M\Omega\ell_\Omega^2,$$

which proves to be the "minimal unit of action per radian". Thus, $\hbar$ is not an

externally imposed constant, but an inevitable consequence of geometric–dynamic coupling. This geometric origin naturally leads to the emergence of typical quantum properties such as angular momentum quantization, the de Broglie relation, the uncertainty principle, and discrete energy levels.

Through a variational derivation of a minimal action functional, the Schrödinger equation

$$i\hbar\partial_t\psi = (-\frac{\hbar^2}{2m}\nabla^2 + V)\psi,$$

is obtained, showing that quantum dynamics is not based on empirical postulates but emerges as the logical necessity of the geometric–action structure.

The logical progression of this paper develops step by step along the path:
"geometric scaling → space–mass coupling → reduced Planck constant → Schrödinger equation"
This forms a closed-loop derivation from geometry to quantum theory. The results indicate that the foundation of quantum mechanics can be naturally grounded in the geometric structure of the Unified Field Theory, providing a robust basis for further interpretation of more complex systems (such as hydrogen, helium, and macroscopic quantum effects).

Supplement to the Abstract: Geometric Framework of the Unified Field Theory

1. Helical Flow of Space

Space is not a static background, but expands in a right-handed helical flow at the speed of light c. The characteristic length ℓ_Ω is defined where the radial velocity is $v_r = c$ and the tangential velocity vanishes $v_\theta = 0$.

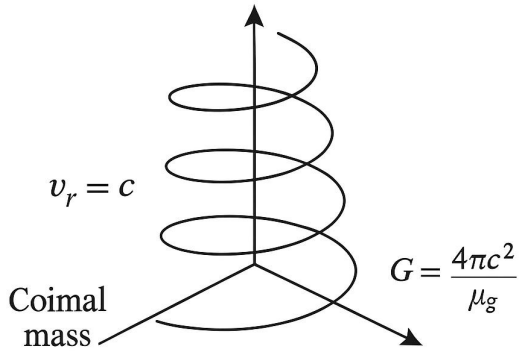

Figure 1. Helical flow of space. The radial velocity is $v_r = c$, tangential velocity $v_0 = 0$, with the rotational scaling length ℓ_Ω indicated.

2. Space–Time Equivalence

Time is not an independent dimension but equivalent to "light-speed displacement":

$$1\,\mathrm{s} \equiv c \times 1\,\mathrm{s} = 3 \times 10^8\,\mathrm{m}.$$

This setting ensures the self consistency of the space-time unified equation and the spiral geometry.

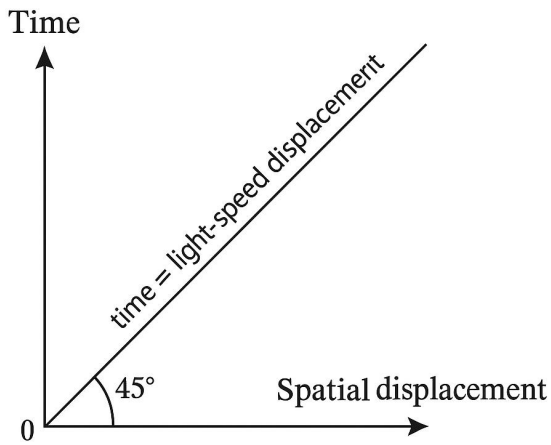

Figure 2. Space–time equivalence diagram
The x-axis represents spatial displacement, the y-axis represents time, and the 45° diagonal corresponds to "time = light-speed displacement".

3. Fundamental Quantities

The framework naturally introduces:

- Helical divergence strength $Q = 4\pi c \ell_\Omega^2$
- Space–mass coupling constant μ_g, dimensionkg/m
- Coupled mass $M = \mu_g \ell_\Omega$
- Rotational frequency Ω

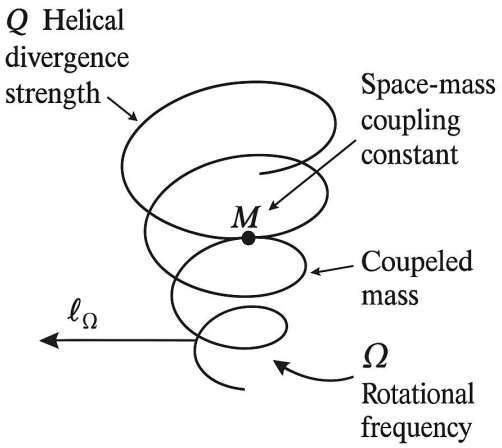

Figure 3. Helical structure with labeled quantities $Q, \mu_g, M, \ell_\Omega, \Omega$.

4. Bridge and Redefinition

From the geometric helical structure and mass coupling one obtains:

$$G = \frac{4\pi c^2}{\mu_g}.$$

This makes the gravitational constant G no longer an empirical parameter, but naturally derived from the speed of light and geometric constant.

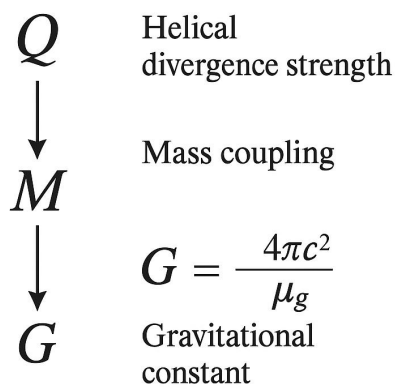

Figure 4. Geometric–dynamical bridging. The chain from Q to M and finally to G.

Part I Geometric Scaling

1.1 Definitions and Assumptions

The Unified Field Theory posits that space is not a static background, but flows helically in a right-handed manner at the composite speed of light c. Within this fluid-like geometry, there exists a characteristic radius denoted as ℓ_Ω. At this location:

- Tangential velocity: $v_\varphi = 0$
- Radial velocity: $v_r = c$

We refer to ℓ_Ω as the rotational scaling length. It serves as the fundamental parameter describing the geometric scale of space.

1.2 Helical Divergence Strength Q

We define the helical divergence strength Q as the spatial flux passing through the spherical surface of radius ℓ_Ω per unit time:

$$(E1) \quad Q = 4\pi c \ell_\Omega^2$$

Dimensional check:

$$[Q] = [c][\ell_\Omega^2] = \frac{m}{s} \cdot m^2 = m^3/s$$

This is consistent with the dimension of volumetric flux.

$$(E1) \quad Q = 4\pi c \ell_\Omega^2$$

This is consistent with the dimension of volumetric flux.

1.3 Conceptual Explanation

Equation (E1) shows that the geometric scaling of space is directly related to the speed of light c.

- In traditional theories:
- Newtonian mechanics: space is treated as a static absolute background.
- Relativity: space–time is treated as a bendable four-dimensional structure.
- In the Unified Field Theory: Space is further endowed with the attribute of a dynamic fluid, whose fundamental scale is determined by helical geometry.

1.4 Experimental and Observational Tests

Ways to test the rationality of Q:

a. Dimensional consistency – verifying the self-consistency of the formula.

b. Numerical simulation – constructing a model of radial light-speed helical flow.

c. Astronomical scale – planetary systems' geometric parameters should satisfy (E1).

1.5 Summary

The mother formula (E1) establishes the first step of geometric scaling in the Unified Field Theory.

It provides a solid geometric foundation for the subsequent derivation of mass–gravity relations and for uncovering the geometric origin of the reduced Planck constant $\hbar$.

Part II Space–Mass Coupling and Gravity

2.1 Definitions and Assumptions

Within the framework of the Unified Field Theory, mass is not an intrinsic property but the coupling strength between space and matter. We define:

$$(E2) \quad M = \mu_g \, \ell_\Omega$$

where:

- M: Coupled mass (kg)
- μ_g: Space–mass coupling constant, with dimension$[\mu_g] = kg/m$;
- ℓ_Ω: Rotational scaling length (m)

Dimensional check:

$$[M] = [\mu_g][\ell_\Omega] = \frac{kg}{m} \cdot m = kg$$

which matches the dimension of mass.

2.2 Mother Formula Derivation

From Part I, the helical divergence strength is:

$$Q = 4\pi c \ell_\Omega^2$$

Assume a geometric–dynamical bridging relation:

$$GM = \frac{c}{\ell_\Omega} Q$$

Substitute Q:

$$GM = \frac{c}{\ell_\Omega} (4\pi c \ell_\Omega^2) = 4\pi c^2 \ell_\Omega$$

Combining with (E2) $M = \mu_g \ell_\Omega$, we can get:

$$(E3) \quad G = \frac{4\pi c^2}{\mu_g}$$

2.3 Conceptual Explanation

- Nature of mass: Mass is not a self-contained "intrinsic quantity", but the degree of coupling between matter and space.
- Gravitational constant G: In the Unified Field Theory, G is not an independent empirical constant but determined by the speed of light c and the coupling constant μ_g.

Comparison:

- Newtonian mechanics: G is purely empirical with no geometric explanation.
- Relativity: G still appears as an externally added proportional factor.
- Unified Field Theory: $G = 4\pi c^2/\mu_g$, fully geometrized.

2.4 Experimental and Observational Tests

1. Planetary and satellite orbits: Infer the consistency of G and μ_g from celestial motions.
2. Multi-system comparison: Test whether μ_g remains stable across different astrophysical systems.
3. Laboratory scale: Measure G with high precision and verify its relation with μ_g.

2.5 Summary

- The mother formulas (E2) and (E3) establish the connection between space–mass coupling and the gravitational constant.
- Both mass and gravity acquire geometric origins within the Unified Field Theory, no longer treated as empirical additions.
- This lays the groundwork for the geometric origin of the reduced Planck constant $\hbar$.

Part III Quantum Scaling

3.1 Derivation of the Reduced Planck Constant $\hbar$ from Geometry

3.1.1 Geometric Preliminaries

Within the Unified Field Theory, space forms a minimal closed loop at the rotational scaling length ℓ_Ω. Previously we obtained:

- Helical divergence strength:

$$(E1) \quad Q = 4\pi c \ell_{\Omega}^{2}, \quad [Q] = m^3/s$$

- Space–mass coupling:

$$(E2) \quad M = \mu_g \ell_{\Omega}, \quad [M] = kg$$

These two mother formulas provide the geometric and mass basis for quantum scaling.

3.1.2 Phase Single-Valuedness and Minimal Action

Let the angular coordinate of the closed loop be $\theta \in [0,2\pi)$。

- Define the action element per radian:

$$\mathrm{d}S = p_\theta \, \mathrm{d}\theta$$

- Action for a full cycle:

$$S_{cycle} = \oint p_\theta \, d\theta = 2\pi p_\theta$$

The requirement of phase single-valuedness is:

$$S_{\mathrm{cycle}} = nh \quad (n \in \mathbb{Z})$$

For the minimal non-zero circulation n=1:

$$p_\theta = \hbar$$

3.1.3 Angular Momentum from Geometry

The Noether quantity corresponding to rotational symmetry is angular momentum:

$$L = I\Omega$$

where the moment of inertia is $I = M\ell_{\Omega}^{2}$. Hence:

$$p_\theta \equiv L = M\Omega\ell_\Omega^2$$

3.1.4 Conclusion: Geometric Origin of the Reduced Planck Constant

By equating the above relations:

$$(E4) \quad \hbar = M\Omega\ell_\Omega^2$$

Dimensional check:

$$[\hbar] = [M][\Omega][\ell_\Omega^2] = (kg)\cdot(1/s)\cdot(m^2) = kg\cdot m^2/s = J\cdot s$$

which is consistent with the reduced Planck constant.

3.1.5 Conceptual Explanation

- Nature of $\hbar$: the minimal unit of action per radian.
- Geometric meaning: it arises from the combination of the minimal closed loop, the rotational scaleΩ , and the coupled mass M.
- Comparison with tradition: in the Unified Field Theory, $\hbar$ is not empirical, but the inevitable product of geometry–symmetry–topology.

3.2 Quantum Properties Derived from $\hbar$

Once $\hbar$ is established as the "minimal unit of action per radian", other quantum laws emerge naturally. Key properties include:

3.2.1 Quantization of Angular Momentum

Phase single-valuedness requires the action around a closed path to be an integer multiple of $2\pi\hbar$:

$$\oint \nabla S \cdot d\mathrm{r} = 2\pi n\hbar$$

This implies angular momentum can only take discrete values:

$$(E4a) \quad L_z = n\hbar, \quad n \in \mathbb{Z}$$

3.2.2 de Broglie Relations

For a plane wave $\psi \sim e^{i(\mathrm{k}\cdot\mathrm{r}-\omega t)}$, the phase gradient yields:

$$\mathrm{p} = \hbar\mathrm{k}, \quad E = \hbar\omega$$

This is the core expression of wave–particle duality.

3.2.3 Commutation Relation and Uncertainty Principle

In operator form:

$$(E4b) \quad [x,\hat{p}] = i\hbar$$

Thus the measurement limit is:

$$\Delta x\, \Delta p \ \geq \ \frac{\hbar}{2}$$

Dimensional check: $[x] = m, [p] = kg \cdot m/s, [\hbar] = kg \cdot m^2/s$, both sides are consistent.

3.2.4 Discreteness of Energy Levels

For the harmonic oscillator, the eigenvalues are:

$$(E4c) \quad E_n = (n + \backslash tfrac12)\hbar\omega, \quad n = 0,1,2,...$$

Here $\hbar$ determines the size of the energy steps.

3.2.5 Geometric Phase

In the presence of a gauge potential $\mathcal{A}$, the phase increment is:

$$\Delta S = \hbar \oint \mathcal{A} \cdot d\ell$$

This reflects the *Aharonov–Bohm* effect and *Berry* phase, showing that$\hbar$ sets the scale of geometric and topological effects.

3.2.6 4π Periodicity of Spin

Quantum states transform as spinor representations under the group SU(2), requiring a 4π rotation to return to their eigenstate.

This stems from the fact that $\hbar$, as the minimal angular momentum unit, corresponds to a double covering scale.

3.2.7 Summary

- Equations (E4a)–(E4c) demonstrate how $\hbar$ universally governs the quantization of angular momentum, momentum, and energy.
- Phenomena such as uncertainty, discrete energy levels, geometric phase, and spin periodicity all follow naturally from the geometric origin of $\hbar$.
- Thus, the core laws of quantum mechanics are no longer empirical postulates but consequences of space geometry and phase single-valuedness.

3.3 Transition from Quantum Scaling to Dynamical Equations

3.3.1 Action Phase and Wavefunction Representation

Since $\hbar$ is the minimal unit of action per radian, the phase of a physical system can be written as:

$$S = \int p_\theta \, d\theta \quad \Rightarrow \quad \psi = \sqrt{\rho}\, e^{iS/\hbar}$$

where:

- $\rho(x,t)$: probability density
- $S(x,t)$: action phase

This representation (often called the Madelung form) serves as the bridge from geometric scaling to the wavefunction.

3.3.2 Continuity Equation and Conservation Laws

- Within the helical flow geometry, space–mass coupling ensures mass conservation.
- From this naturally follows probability conservation:

$$(E4d) \quad \partial_t \rho + \nabla \cdot \left(\rho \frac{\nabla S}{m}\right) = 0$$

This is the continuity equation in quantum mechanics.

3.3.3 Generalized Hamilton–Jacobi Equation

Combining dynamical scaling with minimal action:

$$(E4e) \quad \partial_t S + \frac{|\nabla S|^2}{2m} + V - \frac{\hbar^2}{2m}\frac{\nabla^2\sqrt{\rho}}{\sqrt{\rho}} = 0$$

The first two terms correspond to the classical Hamilton–Jacobi equation, while the last term is the quantum potential introduced by the geometric origin of $\hbar$.

3.3.4 Composite Wavefunction Equation

Combining (E4d) and (E4e) into the wavefunction form $\psi = \sqrt{\rho}\, e^{iS/\hbar}$, the following results are obtained:

$$(E5) \quad i\hbar\partial_t\psi = \left(-\frac{\hbar^2}{2m}\nabla^2 + V\right)\psi$$

This is the Schrödinger equation, not an empirical postulate but the inevitable

outcome of geometric–quantum scaling.

3.3.5 Summary

• Starting from the geometric origin (E4), through phase single-valuedness and wavefunction representation, quantum scaling extends naturally to dynamical constraints.

• (E4d) and (E4e) are the continuity and generalized Hamilton–Jacobi equations, whose synthesis yields (E5), the Schrödinger equation.

• Thus, the fundamental dynamical laws of quantum mechanics are logical necessities derived from spatial geometry and the single-valuedness of action.

Part IV Derivation of the Schrödinger Equation

4.1 Definitions and Assumptions

The Unified Field Theory asserts that quantum wave behavior originates from the minimal unit of action $\hbar$, arising from the helical flow of space.

To derive the dynamical equation, we introduce:

- Probability density: $\rho(x,t)$
- Phase function: $S(x,t)$
- Wavefunction:

$$\psi(x,t) = \sqrt{\rho(x,t)}\, e^{iS(x,t)/\hbar}$$

4.2 Minimal Action Functional

Construct the action functional of the system:

$$(E5a)$$

$$\mathcal{A} = \int d\,t \int d^3\,x\{\rho(\partial_t S + \frac{|\nabla S|^2}{2m} + V)\bullet \quad \frac{\hbar^2}{8m}\frac{|\nabla\rho|^2}{\rho}\}$$

where:

• The first term corresponds to the classical Hamilton–Jacobi action.

• The second term is a quantum potential correction introduced by the geometric scaling of $\hbar$.

Dimensional check:

- $[\rho] = 1/m^3$, $[S] = J \cdot s$, so the dimension of $\rho\partial_t S$ is $(1/m^3) \cdot (J) = J/m^3$, which is consistent with the energy density;
- The dimension of $\hbar^2/(8m) \cdot (|\nabla\rho|^2/\rho)$ is also the energy density.

Thus the overall $\mathcal{A}$ has dimension energy × time, consistent with an action.

4.3 Variational Derivation

1. Variation with respect to S:

yields the continuity equation:

$$(E5b) \quad \partial_t\rho + \nabla \cdot (\rho\frac{\nabla S}{m}) = 0$$

2. Variation with respect to ρ:

yields the generalized Hamilton–Jacobi equation:

$$(E5c)$$

$$\partial_t S + \frac{|\nabla S|^2}{2m} + V\bullet \quad \frac{\hbar^2}{2m}\frac{\nabla^2\sqrt{\rho}}{\sqrt{\rho}} = 0$$

4.4 Composite Wavefunction Equation

Combining (E5b) and (E5c) into the wavefunction form $\psi = \sqrt{\rho}\, e^{iS/\hbar}$, we obtain:

$$(E5) \quad i\hbar\partial_t\psi = \left(-\frac{\hbar^2}{2m}\nabla^2 + V\right)\psi$$

This is precisely the Schrödinger equation.

4.5 Conceptual Explanation

• Non-postulatory nature: Within the Unified Field Theory, the Schrödinger equation is not an empirical postulate, but follows necessarily from the geometric scaling of $\hbar$ and the variational principle of the action functional.

- Physical significance:
- (E5b) ensures probability conservation.
- (E5c) extends the classical Hamilton–Jacobi equation with a quantum potential arising from $\hbar$.
- (E5) unifies them into wavefunction dynamics.

4.6 Experimental and Observational Tests

1. Interference and diffraction: In the electron double-slit experiment, the phase difference corresponds to the difference in action along spatial helices.
2. Atomic energy levels: The hydrogen atom spectrum can be derived from (E5), fully consistent with experimental results.
3. Macroscopic quantum phenomena: Coherence in superconductivity and superfluidity also follows (E5).

4.7 Summary

The mother formula (E5) completes the logical closed loop from geometric scaling to quantum dynamics:

- (E1) Geometric flux
- (E2) Space–mass coupling
- (E4) Reduced Planck constant
- (E5) Schrödinger equation

The Unified Field Theory thus reveals the true root of quantum mechanics:
Geometry → Action → Wavefunction → Dynamics.

Part V Conclusion and Outlook

5.1 Review of the Logical Closed Loop

Within the framework of the Unified Field Theory, this paper followed the principle of "geometry first, conceptual explanation, logical reasoning" and gradually established a complete chain from geometric scaling to quantum dynamics:

1. Geometric scaling:

$$(E1) \quad Q = 4\pi c \ell_\Omega^2$$

The dynamic geometric scale of space is determined by the helical divergence strength Q and the speed of light c.

2. Space–mass coupling and gravity:

$$(E2) \quad M = \mu_g \ell_\Omega, \quad (E3) \quad G = \frac{4\pi c^2}{\mu_g}$$

Mass and gravity are both geometrized, no longer empirical additions.

3. Geometric origin of the reduced Planck constant:

$$(E4) \quad \hbar = M\Omega\ell_{\Omega}^{2}$$

$\hbar$ from the single-valuedness of minimal closed loops and rotational symmetry. Its dimension is self-consistent and its physical meaning explicit.

4. Natural emergence of the Schrödinger equation:

$$(E5) \quad i\hbar\partial_t\psi = \left(-\frac{\hbar^2}{2m}\nabla^2 + V\right)\psi$$

Thus, dynamical laws are no longer empirical postulates but necessary outcomes of geometry and action.

5.2 Conceptual Elevation

• Equivalence of time and space: Spatial geometric flow replaces the static background; time is merely a motion parameter.

• Quantum essence: $\hbar$ is not an externally imposed constant, but the "minimal unit of action per radian", rooted in geometric–topological structure.

• Unification of mechanics: Mass, gravity, and quantum effects all share the same geometric origin, no longer fragmented.

5.3 Directions for Experimental and Observational Tests

1. Planetary and celestial orbits: Verify the geometric relation between μ_g and G.

2. Atomic energy levels and spectra: Derive energy levels from (E5) and compare with experiments.

3. Macroscopic quantum phenomena: Test the geometric–quantum bridge through coherence in superconductivity and superfluidity.

5.4 Outlook

• More complex atomic systems: While the hydrogen atom can already be explained, the geometric framework can be extended to helium and multi-electron systems.

• Cosmological scale: The spatial helical geometry may provide new explanations for dark matter and dark energy.

• Technological applications: If spatial helical geometry can be artificially

manipulated, it could spark revolutionary advances in energy, information, and materials.

5.5 Summary

Quantum mechanics under the unified field theory presents a new picture:

• Geometry → Coupling → Quantum scale → Schrödinger equation

It is a logic closed-loop chain.

• The root of quantum mechanics lies not in empirical assumptions, but in the necessity of spatial geometry and symmetry.

• This has laid a more solid foundation for human science and provided a new way to explore higher-level natural laws.

So far, the five major parts of this paper (abstract, geometric calibration, space mass coupling, quantum scale, derivation of Schrodinger equation, summary and prospect) have been completely closed-loop.

Appendix

A. Table of Main Symbols

Symbol	Meaning	Dimension
c	Speed of light	m/s
ℓ_Ω	Rotational scaling length	m
Q	Spiral divergence strength	m^3/s
μ_g	Space–mass coupling constant	kg/m
M	Coupled mass	kg
G	Gravitational constant	$m^3/(kg \cdot s^2)$

Ω Angular frequency of rotation $1/s$

$\hbar$ Reduced Planck constant (action per radian) $J \cdot s$

h Planck constant (action per full circle) $J \cdot s$

$\rho(x,t)$ Probability density 1/m³

S(x,t) Action phase $J \cdot s$

$\psi(x,t)$ Wave function —

V Potential energy J

m Particle mass kg

B. Master Equations and Dimensional Summary

1. Geometric Scaling

$$(E1) \quad Q = 4\pi c \ell_\Omega^2, \quad [Q] = m^3/s$$

2. Space–Mass Coupling

$$(E2) \quad M = \mu_g \ell_\Omega, \quad [M] = kg$$

3. Gravitational Constant

$$(E3) \quad G = \frac{4\pi c^2}{\mu_g}, \quad [G] = m^3/(kg \cdot s^2)$$

4. Reduced Planck Quantity

$$(E4) \quad \hbar = M\Omega \ell_\Omega^2, \quad [\hbar] = J \cdot s$$

5. Schrödinger Equation

$$(E5) \quad i\hbar\partial_t\psi = (-\frac{\hbar^2}{2m}\nabla^2 + V)\psi$$

References:

1. 基于电磁变化产生的引力场及物体运动　作者：张祥前，徐玉川

文章发表于：2025 年第 1 期（新疆钢铁期刊）

2. 基于正电荷加速运动产生正反引力场的实验　作者：张祥前，徐玉川

文章发表于：2024 年第 16 期（中国图片期刊）

3. 张祥前统一场论相关论文与资料

Zhang,X.(2025).Unified Field Theory （Academic Edition） Hope Grace Publishing

4. Newton, I. Philosophiæ Naturalis Principia Mathematica, 1687.

5. Einstein, A. Die Grundlage der allgemeinen Relativitätstheorie, Annalen der Physik, 1916.

6. Schrödinger, E. Quantisierung als Eigenwertproblem, Annalen der Physik, 1926.

7. Planck, M. On the Law of Distribution of Energy in the Normal Spectrum, Annalen der Physik, 1901.

8. Bohm, D. A Suggested Interpretation of the Quantum Theory in Terms of Hidden Variables, Physical Review, 1952.